

**26<sup>TH</sup> ANNIVERSARY OF  
PCI-IOSS**

**SPINE SURGERY SUMMIT**

in conjunction with

**ISASS AP 2024**



INTERNATIONAL SOCIETY FOR THE ADVANCEMENT OF SPINE SURGERY ASIA-PACIFIC SECTION

28 - 31 AUGUST 2024 | YOGYAKARTA, INDONESIA

# Morphology of retroperitoneal vascular corridor for ATP access to L2-S1 intervertebral disc: MRI study with clinical correlation

Fatmawati General Hospital  
Jakarta - Indonesia



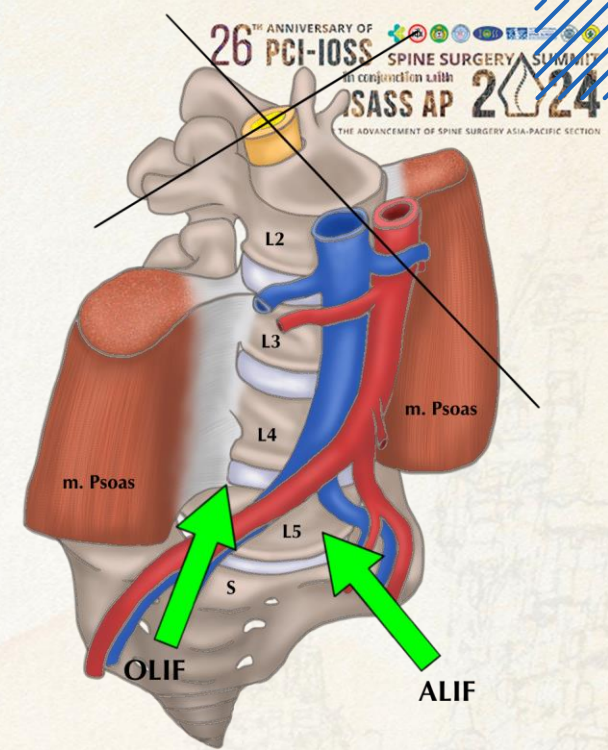
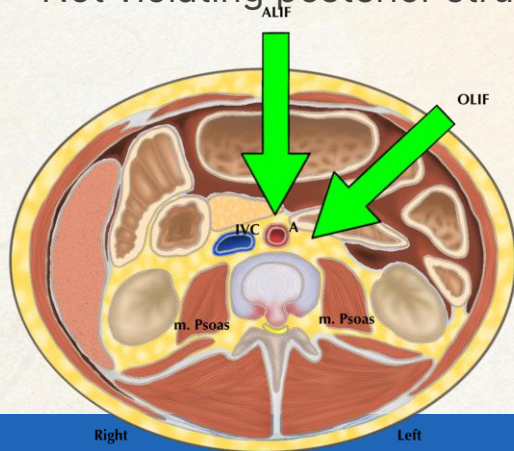
**WWW.PCI-IOSS-ISASSAP2024.COM**

# Disclosure

- I have none to disclose

# Lumbar Interbody Fusion

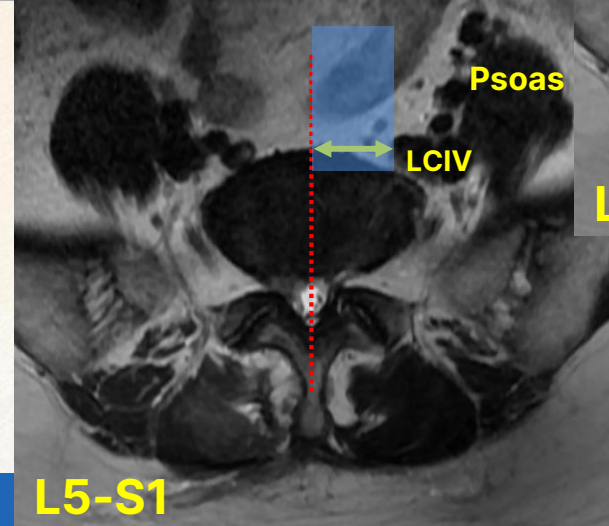
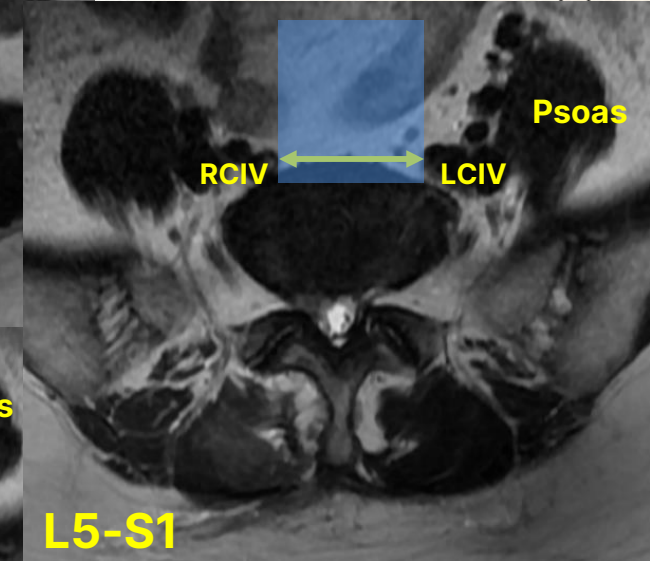
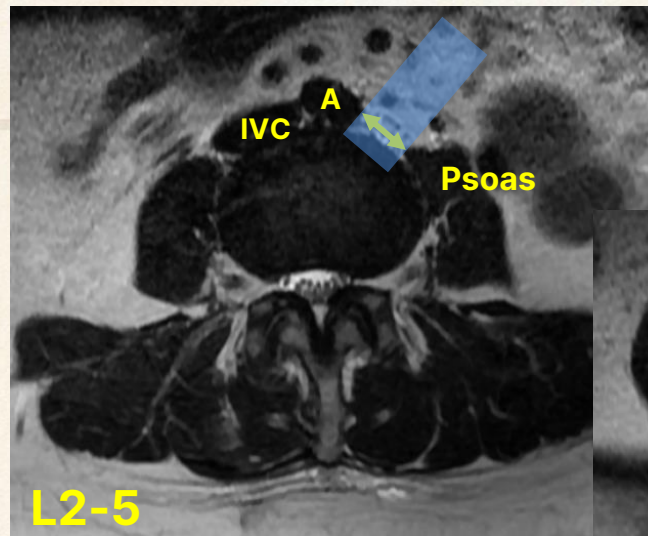
- Anterior/Oblique approach LIF gained popularity
  - Adequate endplate preparation + Larger cage = Better fusion
  - Stronger correction power
  - Indirect decompression
  - Not violating posterior structure



- Risk of vascular injury → OLIF 1,6-4,3%; ALIF 3% → overall 1,9%
- Approach related → fail to clearly identify and mobilize major vessel safely

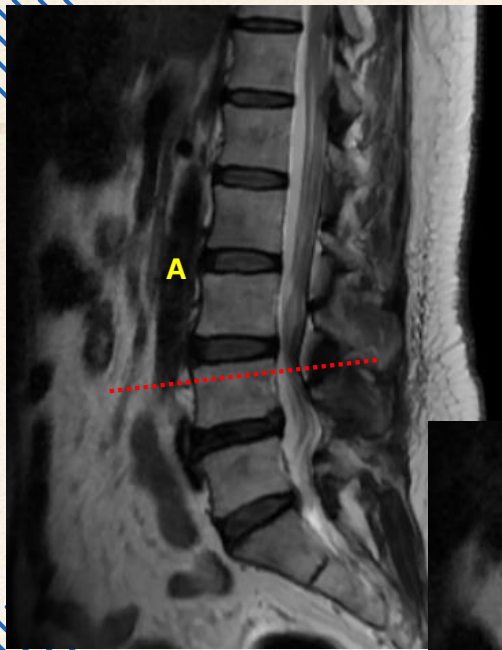
# Method

- 28 - 31 AUGUST 2024 in YOGYAKARTA - INDONESIA
- 122 lumbar MRI in supine; sagittal and axial T2WI
- Criteria:
  - LBP/sciatica patients
  - Degenerative disc/Mild spondylolisthesis
  - No deformity/scoliosis
  - No tumor/infection
  - No previous surgery/instrumentation
- ANTERIOR and OBLIQUE corridor L2-S1 disc

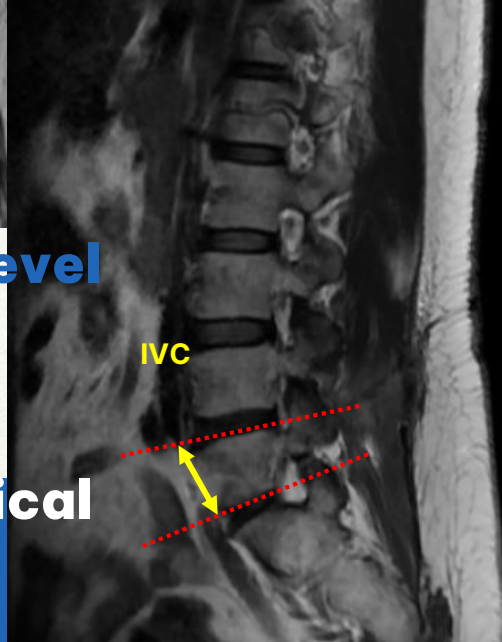
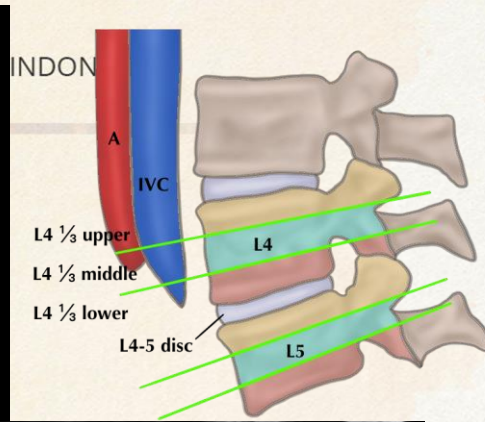


## 1. Oblique Corridor

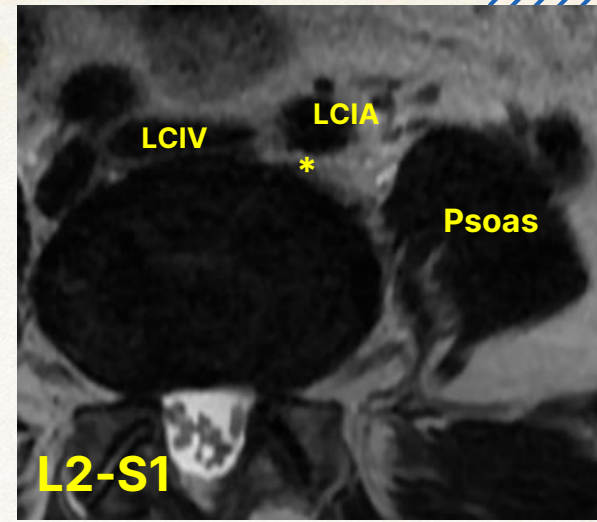
## 2. Anterior Horizontal Corridor



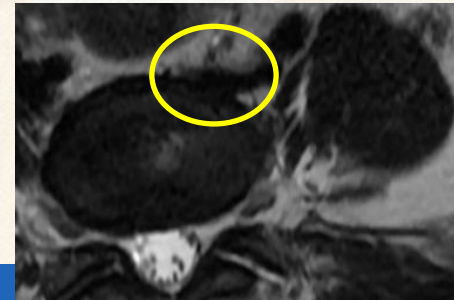
### 3. Bifurcatio Level



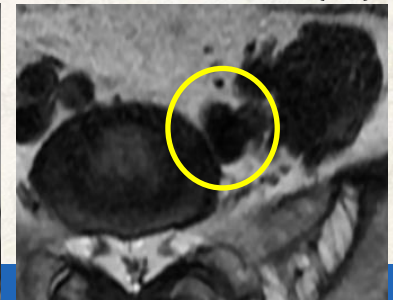
### 4. Anterior Vertical Corridor



### Fat plane (-)

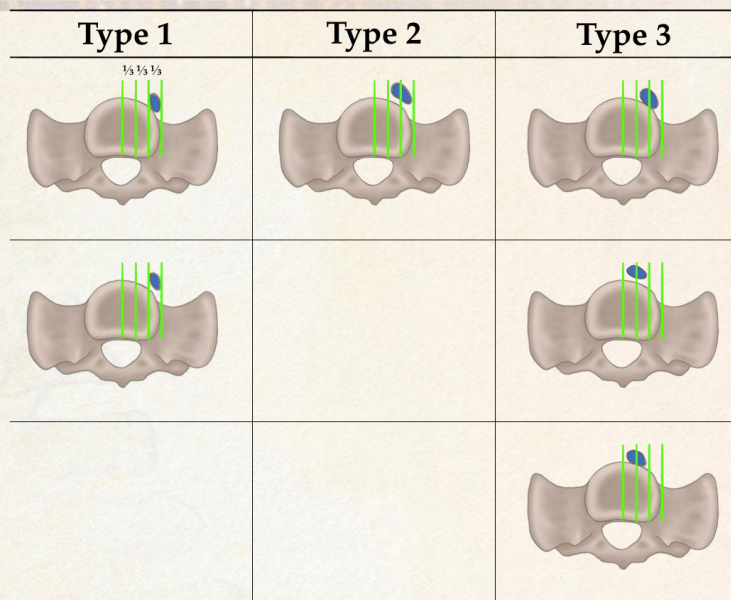


### Fat plane (+)

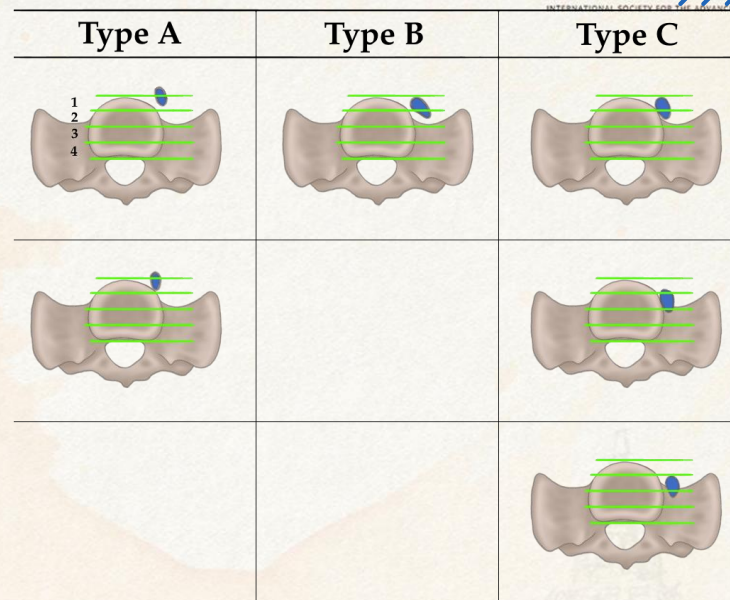


### 5. Perivascular Fat Plane

## 6. LCIV configuration at L5-S1 disc

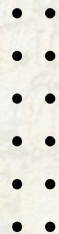


**Coronal Plane<sup>1</sup>**



**Sagittal Plane<sup>2</sup>**

- To determine vessel position and difficulty for mobilization

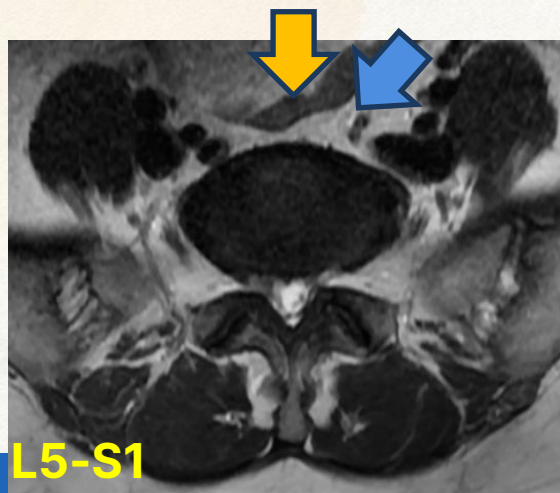


1. Chung NS, Jeon CH, Lee HD, Kweon HJ. Preoperative evaluation of left common iliac vein in oblique lateral interbody fusion at L5-S1. Eur Spine J. 2017;26:2797-2803  
 2. Moro T, Kikuchi S, Konno S, Yaginuma H. An anatomic study of the lumbar plexus with respect to retroperitoneal endoscopic surgery. Spine (Phila Pa 1976). 2003;28:423-428 [discussion: 427-428].

# Surgical Access to L2-S1 Disc



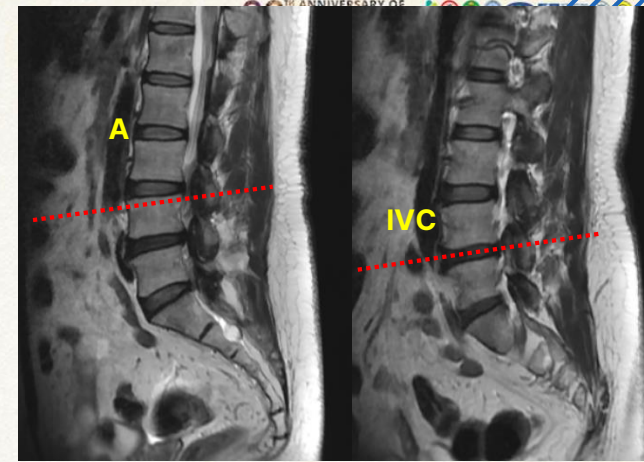
| Disc Level       | Mean distance (mm) ± SD | Min (mm) | Max (mm) |
|------------------|-------------------------|----------|----------|
| L2-3             | 18,66 +/- 6,21          | 3,5      | 39,6     |
| L3-4             | 19,12 +/- 6,8           | 4        | 36,6     |
| L4-5 (Obl)       | 14,53 +/- 7,65          | 0 (3,3%) | 34,2     |
| L5-S1 (Obl)      | 16,46 +/- 7,02          | 0 (4,1%) | 32,7     |
| L5-S1 (Ant)      | 33,47 +/- 10,35         | 0 (1,6%) | 53,4     |
| L5-S1 (Vert/Sag) | 26,08 +/- 10,66         | 0 (0,8%) | 47,5     |



- 91,2% has surgical corridor
- 4 pt no OLIF corr L4-5; 2 pt no ALIF corr L5-S1
- No difference between age or gender
- 38 (31%) OLIF and 21 (17%) ALIF  
→ no vascular injury

# Bifurcatio morphology

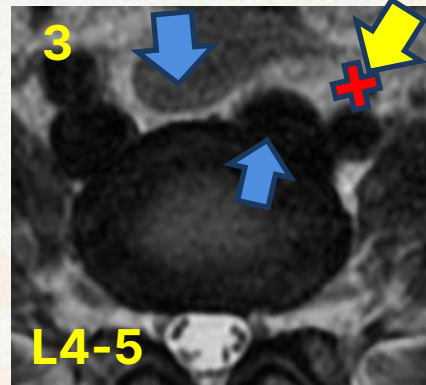
| Level      |                | Aortic Bifurcation |    |      |      | Iliocaval Junction |    |      |      |
|------------|----------------|--------------------|----|------|------|--------------------|----|------|------|
|            |                | Σ                  |    | %    |      | Σ                  |    | %    |      |
|            |                | M                  | F  | M    | F    | M                  | F  | M    | F    |
| L3 body    | Inferior third | NA                 | 1  | NA   | 0,8  | NA                 | NA | NA   | NA   |
| L3-4 disc  |                | NA                 | 3  | NA   | 2,5  | NA                 | NA | NA   | NA   |
| L4 body    | Superior third | 5                  | 9  | 4,1  | 7,4  | 1                  | 1  | 0,8  | 0,8  |
|            | Middle third   | 8                  | 20 | 6,6  | 16,4 | 1                  | 4  | 0,8  | 3,3  |
|            | Inferior third | 11                 | 19 | 9    | 15,6 | 4                  | 16 | 3,3  | 13,1 |
| L4-5 disc  |                | 14                 | 24 | 11,5 | 19,7 | 16                 | 27 | 13,1 | 22,1 |
| L5 body    | Superior third | 2                  | 5  | 1,6  | 4,1  | 9                  | 22 | 7,4  | 18   |
|            | Middle third   | NA                 | 1  | NA   | 0,8  | 6                  | 6  | 4,9  | 4,9  |
|            | Inferior third | NA                 | NA | NA   | NA   | 3                  | 5  | 2,5  | 4,1  |
| L5-S1 disc |                | NA                 | NA | NA   | NA   | NA                 | 1  | NA   | 0,8  |
| Total      |                | 40                 | 82 | 100  | 100  | 40                 | 82 | 100  | 100  |



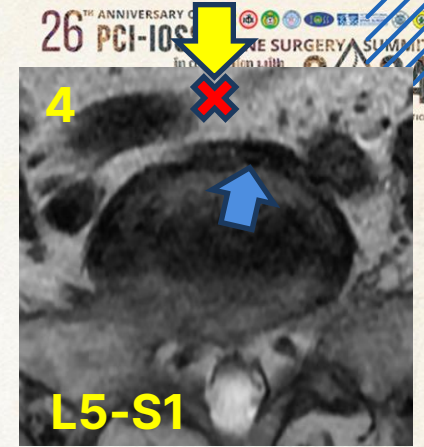
- L4-5 disc as most common junction location
- No difference with age or gender
- Bifurcatio almost always higher than vein junction
- Lower bifurcation = smaller anterior vertical corr

# Presence of Fat Plane

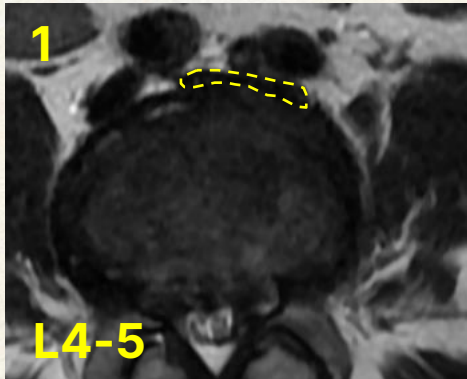
| Disc Level | Mean distance (mm) $\pm$ SD | Min (mm) | Max (mm) |
|------------|-----------------------------|----------|----------|
| L2-3       | 2,12 $\pm$ 1,35             | 0,5      | 9,6      |
| L3-4       | 2,43 $\pm$ 1,57             | 0,3      | 9,9      |
| L4-5       | 3,01 $\pm$ 2,4              | 0 (2,5%) | 11,5     |
| L5-S1      | 2,3 $\pm$ 2,17              | 0 (9,8%) | 12,4     |



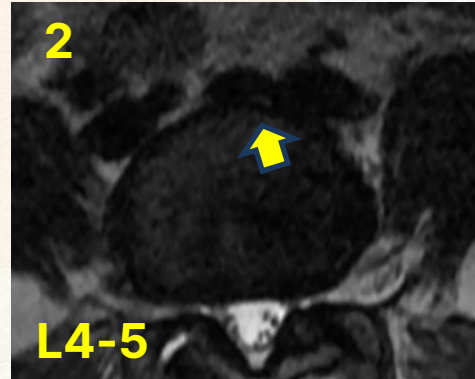
**No obl corr or fat line**



**No corr or fat line**



**No fat line on LCIV**

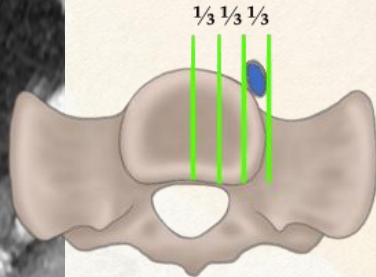
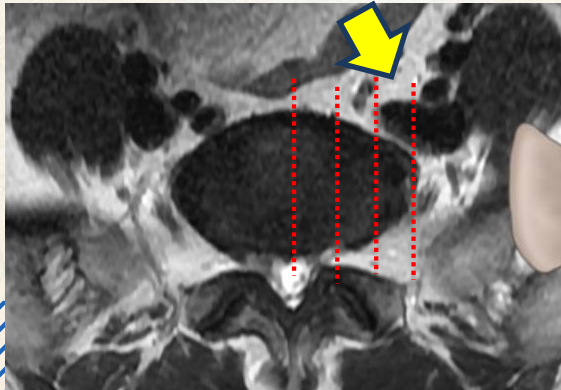


**No corr or fat line**

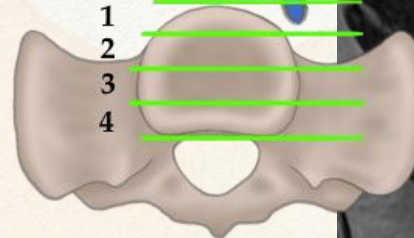
- 3 pt no fat on L4-5 vessel; 12 pt no fat on L5-S1 vessel
- Fat = space for mobilization
- One of major predictors for difficulty in mobilizing vessel for disc access<sup>3</sup>

# LCIV classification

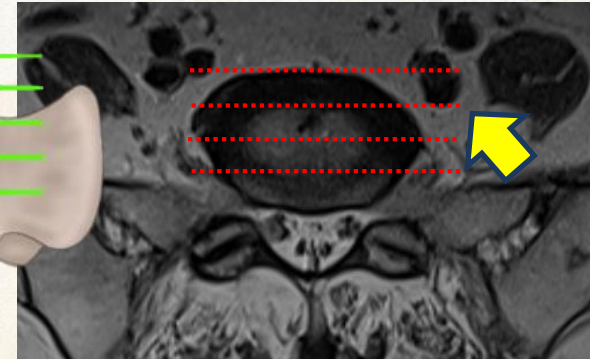
- Most common is type 1 (54,9%) and type A (65,6%)
- LCIV most commonly injured (ALIF/OLIF L5-S1) → position was very related to level of ilioacaval junction
- Type 3 LCIV is the most prone for vessel injury



**Type 1**



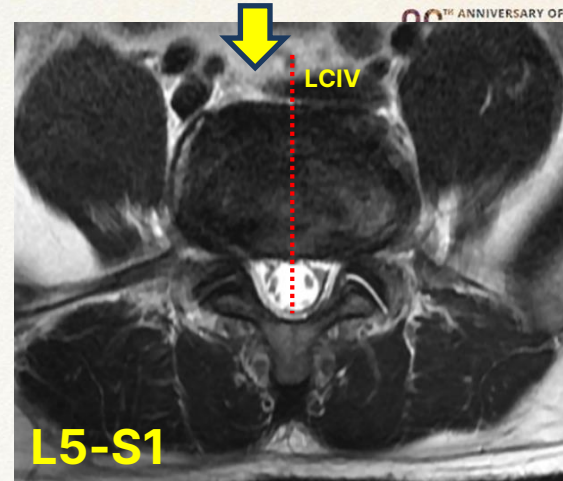
**Type A**



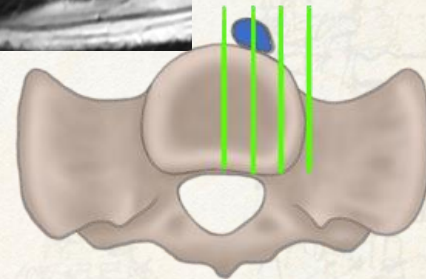
| LCIV position  |        | Σ  |    | %    |      |
|----------------|--------|----|----|------|------|
|                |        | M  | F  | M    | F    |
| Coronal plane  | Type 1 | 20 | 47 | 16,4 | 38,5 |
|                | Type 2 | 9  | 15 | 7,4  | 12,3 |
|                | Type 3 | 11 | 20 | 9    | 16,4 |
| Sagittal plane | Type A | 30 | 50 | 24,6 | 41   |
|                | Type B | 8  | 26 | 6,6  | 21,3 |
|                | Type C | 2  | 6  | 1,6  | 4,9  |

# ALIF L5-S1 in cross midline vessel

- M, 45yo, Spondylolisthesis
- L5-S1 Type 3 LCIV (14,3mm anterior corr)
- ALIF approach from left side
- Fat plane provide access to mobilize LCIV
- No vascular injury



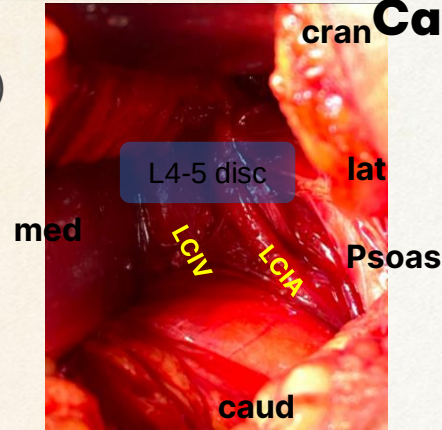
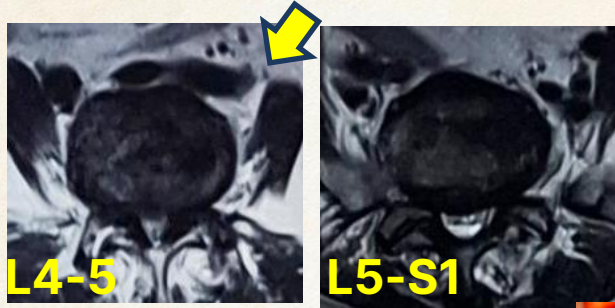
Type 3



# ALIF L4-5 L5-S1

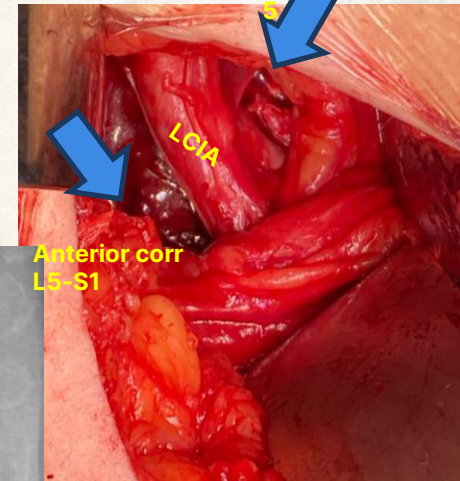
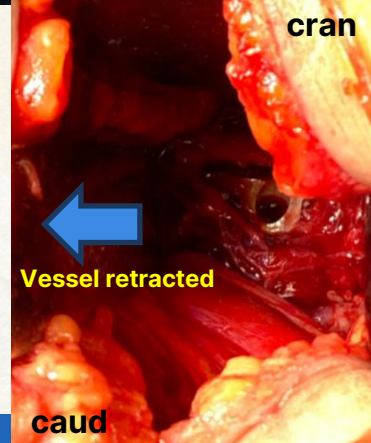
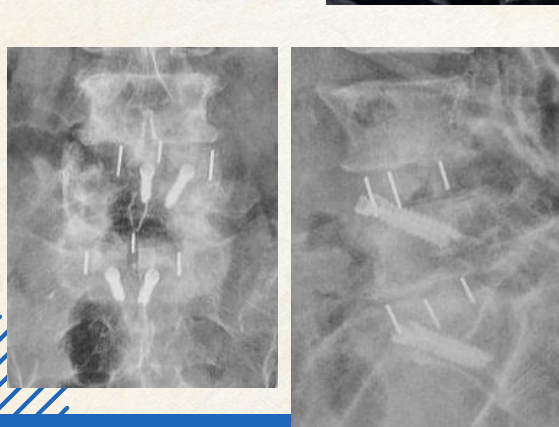
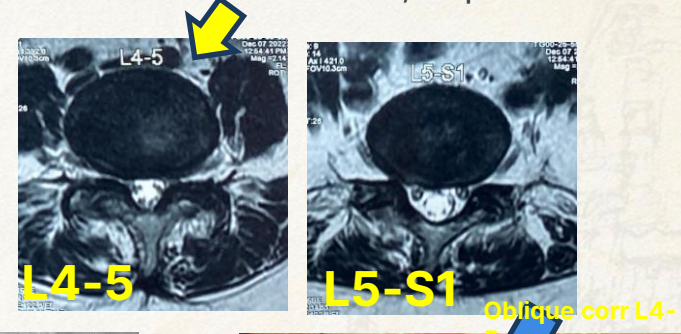
## Case 1

- F, 60yo, Spondylolisthesis
- Small L4-5 corridor, fat plane (+)
- ALIF L4-5 with obl corr, supine



## Case 2

- F, 57yo, Spondylolisthesis
- L4-5 Obl corr (+), fat plane (+)
- ALIF L4-5 via obl corr, supine



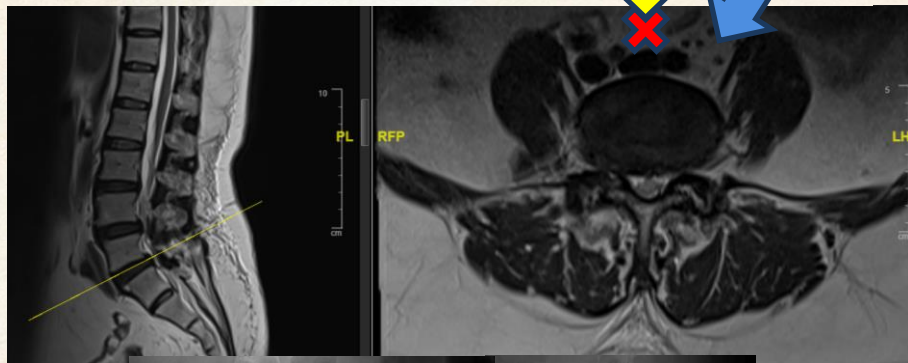
# Lumbosacral Transitional Vertebra (LSTV)

- 7 sacralization (5,7%); 2 lumbarization (1,6%)
- 5 (71%) with access difficulties:
  - No OLIF corr L4-5
  - No ALIF corr L5-S1
  - Very low iliocaval junction (L5 low 1/3 or L5-S1 disc)

## Case 1

F, 56yo, Spondylolisthesis with L5 sacralization

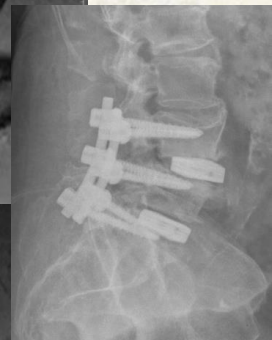
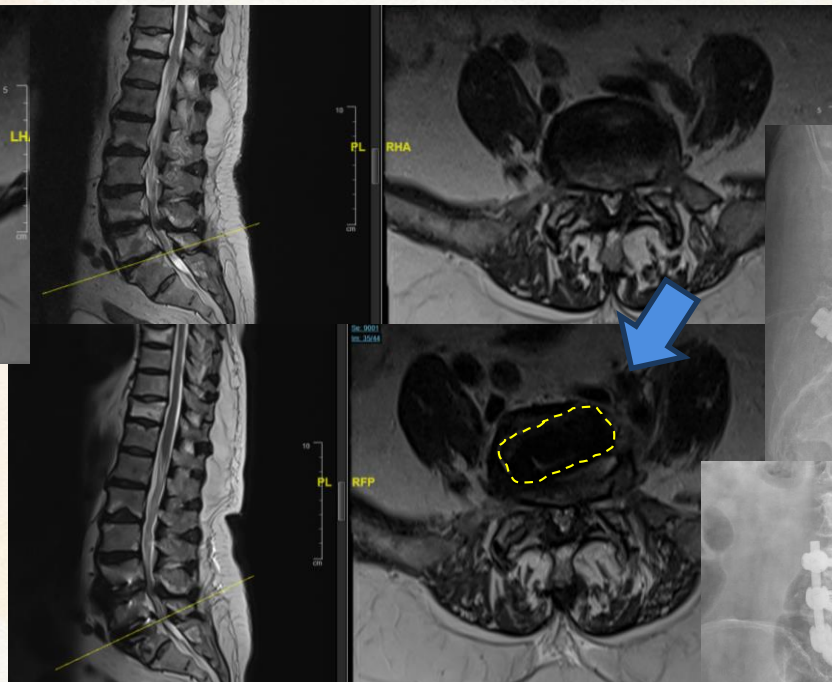
- No ALIF corr



## Case 2

M, 73yo, DDD with S1 lumbarization

- No 5-1 obl corr but noticeable fat plane
- 2-lvl OLIF single position



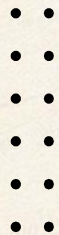
# Take Home Message

- Determining factors for surgical access difficulties:

- Size of surgical corridor
- Presence of fat plane
- LCIV position
- Presence of LSTV

- MRI-based preoperative planning is recommended for anterior/oblique LIF surgery
- Evaluate difficulty determining factors to plan your surgical strategy
- Choose your case wisely based on your learning curve

# Thank You



Erwin Ardian, MD  
Spine Division, Fatmawati General Hospital  
Jakarta, Indonesia

